

NEKHA BOSE

Senior Software Engineer • Full-Stack • AI & Agentic Systems • Cloud Architecture

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PROFESSIONAL SUMMARY

High-impact Senior Software Engineer with 6+ years of experience delivering enterprise-grade, scalable full-stack applications for Fortune 100 clients and high-growth startups. Expert in AI/agentic systems, RAG architectures, LLM integration, and cloud-native development across Azure, AWS and GCP. Microsoft Certified Azure AI Engineer Associate who has led cross-functional teams of 15+ engineers, and consistently driven measurable outcomes from 40% faster ETL pipelines to 12% user engagement lifts. Combines deep technical mastery with strategic product thinking to build software that scales.

CORE COMPETENCIES

Full-Stack (React.js, React Native, Node.js, Java Spring Boot, Python/FastAPI) • TypeScript / JavaScript • RESTful & GraphQL APIs • Microservices
AI Agents & Orchestration (LangChain, LangGraph, LlamaIndex) • RAG Architectures • LLMs (GPT-4o, Claude 3.5, Gemini, Groq, Mistral)
AWS (EC2, Lambda, Glue, S3) • GCP (Compute Engine, Cloud Run) • Docker / Kubernetes / CI/CD • PostgreSQL / MongoDB • ETL / Airflow • Agile Leadership

PROFESSIONAL EXPERIENCE

Lead Software Developer | **BreathelT Inc.** *December 2024 – Present*

- Architected and shipped an AI-powered lifestyle platform end to end in under 6 months: **Node.js** and **TypeScript** backend, **React.js** business portal, and React Native iOS and Android app.
- Built AI recommendation services in **Python FastAPI** across **7 life disciplines** (health, finance, personal branding, relationships, recreation, peace of mind, and citizen responsibility) using OpenAI and Groq LLM APIs plus structured scoring from user signals.
- Integrated external APIs to enrich recommendations and discovery, including Google Maps (location based services), YouTube (video suggestions), and Amazon (product recommendations).
- Designed **75+ PostgreSQL schemas** and delivered **150+ production APIs** on GCP, deployed with Docker and Nginx with Redis based rate limiting for reliability at scale.
- Led and **mentored 15+ engineers** and drove delivery through Agile and Scrum, achieving **30%** faster feature rollout and a **12%** lift in engagement via A/B tested UI improvements.

Senior Software Engineer | **Infosys Limited** *March 2021 – December 2023*

- Delivered enterprise software solutions for **Fortune 100** clients across retail, logistics, and consumer goods sectors, including a flagship engagement with Nike on data infrastructure modernization.
- Led development of a **Java Spring Boot + React.js** e-commerce platform scaled to **10M+** registered users using Docker and Kubernetes, directly contributing to a 20% increase in client's user retention and satisfaction scores.
- Optimized **MongoDB** and **PostgreSQL** integrations to reduce query response times by 40%, improving system throughput for high-volume transactional workloads.
- Spearheaded ETL pipeline redesign for Nike using **Autosys, Apache Airflow, and AWS**, cutting data processing time by **40%** and boosting retrieval speed by 30%, enabling real-time business intelligence.
- Built Tableau dashboards for executive stakeholders, improving decision-making efficiency by **30%** and cross-team productivity by 25% through actionable, real-time analytics.

Software Engineer | **Freelancer** *October 2018 – March 2021*

- Worked with multiple companies and local vendors including Frito, Cogniz Infotech using **Java, C++, React.js, JavaScript, HTML, and CSS** to deliver production ready features end to end across frontend, backend, and integrations.
- Led projects from discovery to deployment, partnering directly with **founders and business owners** to translate goals into reliable, scalable software and iterate fast based on feedback.

KEY PROJECTS

Enterprise AI Learning Management System (RAG + Agentic Platform)

- Designed and built an **enterprise-grade AI-powered LMS** leveraging advanced RAG, enabling natural language Q&A across textbooks, lecture videos, YouTube content, and multi-format course materials - capability to deploy across **multi-tenant** environments with strict data isolation.
- Architected a **fully pluggable RAG stack**: configurable chunking (semantic, page-based, overlap, parent-child), retrieval strategies (BM25, semantic, hybrid), reranking, and support for **4+ LLM** providers (OpenAI GPT-4o, Claude 3.5, Gemini, Groq).
- Delivered agentic orchestration with **Custom, LangChain, LangGraph**, and **LlamaIndex** workflows; integrated FAISS, ChromaDB, Pinecone, and Qdrant vector stores; implemented RBAC and multi-tenant isolation for enterprise compliance.
- Full-stack: **React/TypeScript frontend, Node.js + FastAPI** backend; building enterprise connectors for Google Drive, S3, Azure Blob, Salesforce, and native LMS platforms.

SmartMart - Full-Stack Role-Based E-Commerce Platform

- Built a multi-user e-commerce app with Jakarta EE, JSF/Facelets, JPA ORM, MVC/DAO architecture, Bootstrap-responsive UI, Jakarta Security auth, and Payara Server deployment.

TECHNICAL SKILLS

Frontend: React.js, React Native, Next.js, Vue 3, Angular, Redux, TypeScript, JavaScript (ES6+), HTML5, CSS3, SCSS, D3.js, Bootstrap, FlutterFlow

Backend & APIs: Node.js/Express, Python, FastAPI, Java (Spring Boot), REST APIs, GraphQL, Microservices, Serverless (AWS Lambda), Maven, OpenAPI, OAuth, NPM

AI / Agentic Systems: LangChain, LangGraph, LlamaIndex, RAG (BM25+Semantic+Hybrid), Prompt Engineering, OpenAI GPT-4o, Claude 3.5 Sonnet, Gemini, Groq, Mistral, Llama 3, Azure OpenAI, Embedding Models, Reranking, Vector Stores (FAISS, ChromaDB, Pinecone, Qdrant), AI Agents, Multi-Agent Systems, Tool Use / Function Calling

Cloud & DevOps: AWS (EC2, Lambda, Glue, S3), GCP (Compute Engine, App Engine, Cloud Run), Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions), Nginx, PM2, Git

Databases: PostgreSQL, MySQL, SQLite, MongoDB, Amazon S3, Google Cloud Storage, Teradata

Data & ETL: Apache Airflow, Autosys, AWS Glue, Kafka, Tableau, ETL Pipeline Design, Real-time Data Processing

Testing & Methodologies: A/B Testing, Postman, Swagger, JUnit, Jasmine, TDD, Agile/Scrum, Kanban, SDLC, Webpack, Babel

Design & Collaboration: Figma, Sketch, Jira, Confluence, InVision, Slack, UI/UX Implementation, Cross-functional Communication

EDUCATION

M.S. Information Technology and Management - GPA: 3.90 / 4.0

Illinois Institute of Technology, Chicago, IL

B.Tech. Information Technology

APJ Abdul Kalam Technological University, Kerala

CERTIFICATIONS & ACHIEVEMENTS

- Microsoft Certified: Azure AI Engineer Associate - Azure AI services, cognitive services, and cloud-based AI solution architecture
- Best Employee of the year at BreathelT Inc.
- Best Student Project Award - Braille Converter, APJ Abdul Kalam Technological University
- Research Paper Presenter - SERB-Sponsored National Conference on Computational Methods for Renewable Energy